Strategic Intelligence & Foresight Assessment: El Niño as a Threat Multiplier in Southeast Asia and Oceania (2026–2040)

Estimative-language convention: "Almost certain" (⩾90%), "Likely" (60–85%), "Roughly even" (~50%), "Unlikely" (15–35%), "Remote" (⩽10%). Confidence (low/moderate/high) reflects source quality and corroboration.

TL;DR

- **An El Niño is now operative.** NOAA issued an El Niño Advisory in June 2026 with a forecast ~63% chance of a "very strong" event (SST anomaly >2.0°C) by autumn, and a positive Indian Ocean Dipole is likely developing in parallel — a combination that historically maximizes drought and fire across maritime Southeast Asia. We assess (high confidence) this will act as a *threat multiplier* on pre-existing structural vulnerabilities rather than as an independent shock.
- **The most dangerous dynamic is compounding cross-sector failure**, not any single hazard: drought → hydropower/agriculture/water failure → inflation and fiscal stress → governance strain, layered onto record Chinese debt-repayment pressure, maritime-chokepoint fragility, and great-power competition. The most exposed states are Laos, Myanmar, the Philippines, Timor-Leste, Cambodia, the Pacific atoll states, and (agriculturally) Vietnam and Thailand.
- **For the United States, the principal risks are second-order:** disruption to semiconductor back-end and critical-mineral supply chains running through Malaysia/Indonesia, simultaneous HADR demands that compete with deterrence posture, and Chinese exploitation of crisis-driven dependency (water diplomacy on the Mekong, emergency financing, security pacts). The central opportunity is to convert disaster-response and infrastructure-resilience leadership into durable strategic advantage.

Key Findings

1. The trigger is real and near-term. As of mid-2026 the equatorial Pacific transitioned to El Niño, with the IRI/CPC plume assigning ~97–98% probability to El Niño persistence through the 2026–27 boreal winter and NOAA estimating a 63% chance of a "very strong" event. El Niño's regional impacts peak in the second boreal winter after onset, so the worst regional effects are likely to fall in the **dry season of 2027** (roughly Dec 2026–May 2027). This gives governments a ~6–12 month warning window.

2. The 2023–24 analog shows the transmission mechanism. The last El Niño (2023–24) was among the five strongest on record. It produced: Philippine farm losses (131 areas placed under a state of calamity and ~2.1 million people affected); Thai rice output cut ~5.6–6%; Mekong Delta saltwater intrusion damaging crops and leaving ~73,900 households short of water; Indonesian drought and fire risk concentrated in Java, Sumatra, and Kalimantan. Academic work quantifies the long tail: per Callahan & Mankin (*Science*, 2023), "$4.1 trillion and $5.7 trillion in global income losses" were attributable to the 1982–83 and 1997–98 El Niño events respectively, with downturns persisting "as long as 14 years"; Cai et al. (*Nature Communications*, 2023) reach similar multi-trillion-dollar conclusions. Indonesia's cumulative per-capita GDP loss has been estimated at ~13% relative to a no-El Niño counterfactual (NTU/CityU Hong Kong). Vietnam News + 2

3. Water is the master variable, and the Mekong is the geopolitical fault line. China's upstream dams (Stimson Mekong Dam Monitor) restrict wet-season flow and release in dry season, decoupling downstream water availability from Beijing's discretion. During El Niño dry phases, Chinese reservoir operations become *de facto* regional water diplomacy: in 2023 China's Xiaowan reservoir filled to record levels while the lower Mekong hit record lows. Vietnam's Mekong Delta — ~18 million people, ~12% of national GDP, the national rice bowl — faces saltwater intrusion losses estimated at ~US$2.7 billion/year and could lose ~10% of dry-season rice area to salinity by 2050. Vietnam News + 2

4. Energy systems fail precisely when demand peaks. El Niño heat drives air-conditioning demand up while drought cuts hydropower — a structural squeeze. In April 2024 the Philippines saw dozens of plants on forced outage, repeated yellow/red alerts on the Luzon and Visayas grids, and rotational cuts in the provinces. The Philippine power system's vulnerabilities are "structural, not seasonal" (ICSC), and the country is now exposed to global LNG price swings. Laos, the would-be "Battery of Southeast Asia," has hydropower revenues undermined by both drought and chronic overcapacity/debt.

5. Food-price shocks transmit regionally and globally. Asia produces and consumes ~90% of the world's rice. El Niño years have historically preceded rice-price inflation; India's 2023 export curbs plus El Niño pushed Thai benchmark rice above $650/ton. The risk is a 2007–08-style cascade if exporters (Thailand, Vietnam) impose restrictions. The US imports over half its rice from Southeast Asia, creating direct exposure.

6. Debt and fiscal fragility convert weather shocks into solvency shocks. Per the Lowy Institute's May 2025 *Peak Repayment* report (Riley Duke), developing countries owe a record $35 billion in 2025 debt repayments to China, "of which $22 billion is set to be paid by 75 of the world's poorest and most vulnerable countries," with Beijing now "more debt collector than banker to the developing world." Laos is in IMF-assessed debt distress, public debt >100% of GDP, surviving only on repeated Chinese deferrals; an El Niño-driven hydropower/revenue shock could trigger default. Cambodia, Myanmar, and Laos are most China-dependent for development finance. Al Jazeera + 3

7. The chokepoints are the global transmission belt. Per the US EIA's *World Oil Transit Chokepoints* (2025), the Strait of Malacca is "the largest oil transit chokepoint in the world" at ~23.7 million b/d in 2023 — roughly 22% of global oil demand and 29% of seaborne oil trade — and China accounted for 48% of import volumes transiting the strait in the first half of 2025, with ~80% of China's crude imports passing through it. Climate disruption (not closure) raises insurance, rerouting, and congestion risk via Lombok/Sunda/Makassar — alternatives that are deeper but longer and themselves Indonesian-controlled. The 2026 US-Indonesia Major Defense Cooperation Partnership has injected new strategic salience into the strait.

8. Critical infrastructure is multiply exposed. Undersea cables (95%+ of internet traffic) face both natural-disaster risk and demonstrated Chinese gray-zone sabotage (⩾11 incidents around Taiwan 2023–2025). The Johor–Singapore data-center boom collides directly with El Niño water/power scarcity. Coastal megacities (Jakarta sinking up to ~15 cm/yr; Manila, Bangkok, Ho Chi Minh City) face subsidence + saltwater intrusion + storm surge.

Details

El Niño × the twelve stressor channels

Water/drought/hydropower. The El Niño–positive IOD combination is the worst case for maritime Southeast Asia, suppressing rainfall over Indonesia, Malaysia, the southern Philippines, and the lower Mekong. Groundwater over-extraction (Jakarta; Manila's reliance on Angat) means drought immediately stresses urban supply. Hydropower-dependent grids (Laos, Vietnam's north, Philippine Mindanao, parts of Malaysia) lose generation as reservoirs draw down.

Agriculture/fisheries/food. Heat stress above ~33°C cuts rice yields; the 2015–16 "Godzilla" El Niño cut regional rice output by ~15 million tonnes and left over 1 million Vietnamese food-insecure. Marine heatwaves (documented in the Singapore Strait/South China Sea in 2023) disrupt fisheries and coral, hitting protein supply and coastal livelihoods. Indonesia's government declared rice self-sufficiency for 2025 (BPS estimate ~34.7 million tonnes, zero public-consumption imports, record ~4-million-tonne Bulog reserves) — but independent analysts (CIPS, CIFOR-ICRAF) dispute the sustainability of the "food estate" model and project a ~4.2-million-tonne structural deficit by 2030. A strong El Niño is the most likely near-term test of that claim. ISEAS-Yusof Ishak Institute...

Energy. Beyond hydropower loss, El Niño raises fuel-import bills (diesel for irrigation pumps and island generators; LNG for peaking) at the moment fiscal space is tightest. Renewable transitions are double-edged: solar performs well in drought-sunshine but hydropower and, in Australia, wind underperform; grid inflexibility is the binding constraint.

Heat/extreme weather/health/fire. 2023–24 brought 40–47°C readings across the region. Fire risk concentrates in Sumatra and Kalimantan peatlands, generating transboundary haze. The 2015 haze episode was catastrophic for health: a Harvard–Columbia study (Koplitz et al., *Environmental Research Letters*, 2016) estimated ~100,300 premature deaths across Indonesia, Malaysia, and Singapore (~91,600 in Indonesia, ~6,500 in Malaysia, ~2,200 in Singapore) — a recurring Indonesia–Singapore–Malaysia diplomatic irritant. Heat amplifies dengue, cholera, and labor-productivity loss. Mighty Earth CNBC

Population/urbanization/migration. ADB projects tens of millions of climate migrants in Asia-Pacific by 2050 (one estimate: 62.9 million even under existing pledges). Coastal megacities concentrate exposure; Laos already exports ~286,000+ documented workers to Thailand amid its economic crisis, a template for distress migration. Internal rural-to-urban displacement strains slum infrastructure.

Supply chains. Malaysia is a global semiconductor back-end (assembly/test/packaging) hub; Penang and the Kuala Lumpur corridor are exposed to grid instability and flooding. Indonesia produced 57% of global refined nickel in 2024 (Fastmarkets) — but C4ADS (2024) found "more than 75 percent of refining capacity in Indonesia is controlled by Chinese stakeholders." Myanmar supplies the majority of China's heavy rare-earth feedstock from Kachin (44,000 tonnes in 2024, ~57% of Beijing's imports) — both subject to El Niño-adjacent disruptions (flooding of mines, hydropower for smelters) and conflict. Fastmarkets American Experiment

Finance/inflation/fiscal. The IMF historically estimated strong El Niños add ~4 percentage points to commodity-price inflation; Bloomberg Economics models ~0.3–0.5 pp GDP drag for affected economies. El Niño's supply-side nature limits central-bank tools, raising stagflation risk and worsening debt-servicing for dollar-denominated borrowers.

Political instability/unrest. Hsiang, Meng & Cane (*Nature*, 2011) found "the probability of new civil conflicts arising throughout the tropics doubles during El Niño years relative to La Niña years" (~3% to ~6%), and that ENSO "may have had a role in 21% of all civil conflicts since 1950." Indonesia's August 2025 protests (≥10 dead, 1,000+ injured, legislative buildings burned after a police vehicle killed a delivery driver) show how fast economic grievance ignites; layering food/energy inflation onto such tinder is destabilizing. Myanmar's civil war (junta controlling ~21% of territory; ~3.5M+ displaced; a March 28, 2025 Mw 7.7 earthquake along the Sagaing Fault killing ~3,600–5,350 and affecting up to ~17–22 million) is the region's worst-case of compounding state failure, further entrenched by the contested December 2025 junta elections. PubMed + 2

Strategic competition. China, the US, Japan, India, and Australia compete for influence via infrastructure finance, security pacts, disaster response, and minerals access. Crises are influence-acquisition opportunities: whoever delivers relief and financing fastest gains leverage.

Military/HADR/maritime. US INDOPACOM and allies (the Philippines, Japan, Australia) increasingly rehearse HADR (Balikatan 25, Doshin-Bayanihan); the November 2025 Philippine typhoon response (Typhoon Kalmaegi) drew US/Japan/Australia support coordinated through INDOPACOM's Center for Excellence in Disaster Management. But simultaneous multi-country disasters could overwhelm response capacity and pull naval assets from deterrence missions — a readiness dilemma.

Chokepoints. The key insight is that climate stress acts through *insurance, congestion, and rerouting* rather than physical closure, with cascading systemic (not just energy) effects given Malacca's embedding in manufacturing supply chains.

Critical infrastructure. Undersea cables, ports (subsidence/storm surge), power grids (heat/drought), water systems (saltwater intrusion), telecoms, and data centers form an interdependent web. The Johor data-center cluster — ~5.8 GW planned capacity, with Malaysia's utility TNB reporting >11 GW of applications (over 40% of Peninsular Malaysia's ~27 GW installed capacity) — is on a collision course with water/power scarcity; Malaysia paused non-AI data-center development and Johor halted new high-water-use approvals in 2025, with residents protesting over water scarcity in early 2026. Singapore's own 2019 data-center moratorium (DCs already ~7% of national electricity) is what diverted demand to Johor.

China's regional influence

China's leverage rises in crises through: (1) **Mekong water diplomacy** — discretionary dam releases as carrot/stick; (2) **emergency financing and debt deferral** — the lever holding Laos from default; (3) **BRI infrastructure** — first-half 2025 saw record BRI engagement (US$66.2B construction + US$57.1B investment), though Southeast Asian completion rates are low (~33% of megaprojects per Lowy) and the region is being displaced by Africa/Central Asia; (4) **critical-mineral control** — refining dominance plus Indonesian nickel and Myanmar rare earths; (5) **security/policing pacts** — Solomon Islands (2022), Cook Islands (2025), outreach to PNG, Vanuatu, Kiribati; (6) **military presence** — naval circumnavigation of Australia and Tasman live-fire (Feb 2025), suspected Cambodia (Ream) access. Constraints: debt-collector backlash, ASEAN hedging (the Philippines and Vietnam restraining Chinese finance), PNG's choice of Australia (October 2025 "Pukpuk Treaty"), and the KIA's seizure of Kachin rare-earth hubs disrupting Beijing's feedstock. Center for Strategic and Int...

US national & economic security

- **Supply chains/semiconductors:** Malaysian back-end exposure; resilience requires friend-shoring and grid-hardening support.
- **Critical minerals:** Chinese dominance of nickel/rare-earth refining is the strategic chokepoint; Myanmar's Kachin instability and the KIA's October 2024 seizure of Chipwi/Pangwa mining hubs disrupt China's feedstock — an opportunity and a wildcard. Stimson Center
- **Trade/shipping:** Malacca/South China Sea disruption raises costs system-wide; US rice import dependence on the region is a direct consumer exposure.
- **Defense posture/alliances:** Treaty commitments to the Philippines (MDT), Thailand, and Australia (ANZUS/AUKUS) plus EDCA basing mean US forces are first responders; HADR competes with deterrence.
- **Economic exposure:** US firms (data centers, semiconductors, energy) are investing heavily (the Philippines Luzon Economic Corridor energy/data-center package; Johor hyperscalers), increasing exposure to regional grid/water failure.

Scenarios (1–15 year horizon)

Scenario 1 — Regional Resilience (likelihood: ~25%, *unlikely-to-roughly-even*). Governments use the warning window for anticipatory action (FAO models), drought-resistant rice, irrigation upgrades, grid diversification, and ASEAN cooperation (no export bans; APTERR reserves deployed). *Leading indicators:* early reservoir management, no rice-export restrictions, functioning ASEAN haze/food coordination, IMF-China cooperation on Lao debt. *Winners:* Singapore, Vietnam (adaptation leader), Australia (food exporter), US (partnership dividends). *Losers:* none catastrophic. *Economic impact:* GDP drag limited to ~0.3–0.5 pp in affected states. *Geopolitics:* status quo competition; ASEAN centrality holds. *US interest:* favorable — resilience partnerships deepen.

Scenario 2 — Chronic Instability (likelihood: ~50%, *roughly even / most likely*). Recurring El Niño/La Niña whiplash produces periodic crises and slow deterioration: episodic blackouts, food-price spikes, haze, localized unrest, creeping debt distress. No single collapse, but cumulative erosion. *Leading indicators:* repeated states of calamity, persistent double-digit food inflation in vulnerable states, Lao-style currency/debt stress spreading, rising distress migration. *Winners:* China (steady leverage via financing/water), adaptation-tech and LNG suppliers. *Losers:* Laos, Myanmar, rural poor, fiscally weak states. *Economic impact:* sustained ~0.5–1.5 pp annual GDP drag in exposed economies; trillions globally over the period. *Geopolitics:* gradual Chinese influence accretion checked by hedging. *US interest:* managed competition; persistent HADR tempo; risk of strategic distraction.

Scenario 3 — Cascading Indo-Pacific Crisis (likelihood: ~25%, *unlikely but high-impact*). A very strong El Niño + positive IOD drives simultaneous drought, crop failure, blackouts, and haze across multiple countries; food-export bans trigger a price spiral; a Lao/Myanmar-type fiscal collapse spreads; unrest topples or destabilizes a government; a maritime incident or cable-sabotage event coincides; mass displacement crosses borders. *Leading indicators:* coordinated export bans, simultaneous multi-grid failures, a sovereign default, large-scale unrest plus a South China Sea/Taiwan-cable incident in the same window. *Winners:* few; possibly China if it positions as crisis financier/relief provider, or the US if it leads response decisively. *Losers:* broad — region-wide recession, humanitarian emergency. *Economic impact:* multi-trillion-dollar global drag; regional contraction. *Geopolitics:* forced realignments; ASEAN cohesion tested to breaking; elevated conflict risk. *US interest:* severe — competing HADR and deterrence demands, supply-chain shock, pressure on alliance credibility.

Recommendations

Stage 1 — Now through the 2027 dry season (warning window):

- **Pre-position HADR and energy:** expand EDCA-site humanitarian stockpiles; stage generators/fuel and water-treatment capacity in the Philippines, Pacific atolls, and Timor-Leste. *Benchmark to escalate:* NOAA/BOM confirmation of a "very strong" event (SST >2.0°C) by October 2026.
- **Lock in food-trade openness:** lead diplomacy to pre-commit ASEAN+3 to no rice-export bans and to activate the ASEAN Plus Three Emergency Rice Reserve (APTERR). *Trigger:* Thai benchmark rice >$600/ton or any new export restriction.

- **Mekong transparency:** fund and amplify the Stimson Mekong Dam Monitor; press China via the Mekong-US Partnership for real-time release data. *Trigger:* lower-Mekong levels at record lows while upstream reservoirs fill.

Stage 2 — Structural resilience (1–5 years):

- Co-finance grid diversification (solar+storage, interconnections) and water reuse with the Philippines, Vietnam, and Malaysia, explicitly tied to data-center/semiconductor resilience.
- Accelerate critical-mineral friend-shoring (nickel processing outside Chinese ownership; allied rare-earth refining) and undersea-cable hardening (advance the Critical Undersea Infrastructure Resilience Initiative; expand the Quad cable partnership; LEO satellite backups).
- Support Lao debt restructuring via IMF-China cooperation to prevent a destabilizing default. *Benchmark:* Lao external-debt-service-to-revenue ratio and reserve coverage (currently ~2 months of imports).

Stage 3 — Contingency (if Scenario 3 indicators trip):

- Stand up a multinational HADR coordination cell (US-Japan-Australia-Quad) with a pre-agreed division of labor so deterrence assets are not stripped.
- Prepare emergency rice/financing facilities to out-compete coercive Chinese crisis lending.

Caveats

- **Forecast uncertainty:** the June 2026 advisory was issued near the spring predictability barrier; peak strength (no category exceeded ~37% probability per NOAA) and timing remain genuinely uncertain. A weaker-than-feared event would shift probability mass toward Scenario 1.
- **El Niño strength ≠ impact:** strong events raise probabilities but do not guarantee severe regional outcomes; local rainfall can deviate (NOAA, ARC Centre for Climate Extremes).
- **Attribution:** much regional damage stems from structural factors (upstream dams, groundwater depletion, debt, governance) that El Niño *amplifies* rather than causes. Disaggregating is difficult.
- **Source caveats:** Indonesian rice self-sufficiency figures are official/state-aligned and disputed by independent analysts; Myanmar casualty and territorial-control figures vary widely by source; data-center capacity figures conflate pipeline with built capacity (Johor built/live capacity is ~487–580 MW versus ~5.8 GW planned).

Black swans / high-impact low-probability outcomes (2026–2040)

- A compound El Niño-drought + South China Sea or Taiwan-Strait military incident in the same season, splitting US attention.

- Coordinated undersea-cable sabotage isolating a Pacific state or Taiwan during a disaster.
- A Mekong "water weaponization" episode during acute drought, triggering an interstate crisis.
- Cascading sovereign defaults (Laos first) spreading contagion to frontier markets.
- A category-defining "Godzilla-plus" El Niño coinciding with a major regional earthquake/volcanic event (the region sits on the Ring of Fire — cf. the March 2025 Myanmar quake).
- Abrupt collapse of the Myanmar junta, releasing refugee flows and contested rare-earth zones.

Most vulnerable countries (ranked)

1. **Laos** (debt distress + hydropower dependence + currency fragility)
2. **Myanmar** (civil war + earthquake + state failure + drought)
3. **Philippines** (grid fragility + agriculture + typhoon/El Niño whiplash + fiscal strain)
4. **Timor-Leste & Pacific atoll states** (existential climate exposure, thin buffers)
5. **Cambodia** (China dependence + Tonle Sap/Mekong reliance)
6. **Vietnam** (Mekong Delta food-security exposure, but stronger adaptation)
7. **Indonesia** (vast exposure but improving buffers; political volatility)
8. **Thailand** (agricultural exposure; better reserves)
9. **Malaysia** (water/power-data-center collision; more diversified)
10. **Singapore/Australia/New Zealand** (most resilient; financial/secondary exposure)

Most critical threats to regional stability (ranked)

1. Compounding food-water-energy failure during peak El Niño
2. Debt/fiscal crises converting weather shocks into solvency shocks
3. Mekong water insecurity as a geopolitical fault line
4. Maritime-chokepoint and undersea-cable fragility
5. Climate-driven displacement and urban strain
6. Governance breakdown/unrest in fragile states
7. Great-power competition crowding out cooperative crisis response

Systemic risk pathways & feedback loops

Core domestic loop: Drought → (hydropower loss + crop failure + water scarcity) → (energy inflation + food inflation) → fiscal stress + household distress → political unrest + distress migration → governance weakening → reduced adaptive capacity → greater vulnerability to the *next* shock. **Parallel external loop:** Fiscal distress → demand for Chinese emergency finance/water cooperation → increased dependency → reduced strategic autonomy → constrained ability to diversify → deeper structural vulnerability. **Amplifiers:** chokepoint insurance/rerouting costs; cable/grid cascading failures; export-ban contagion.

Transition indicators (manageable stress → systemic crisis)

- Two or more major exporters impose rice-export bans within one season.
- Simultaneous multi-country grid emergencies during peak demand.
- A sovereign default (Laos as first domino) plus currency contagion.
- States of calamity declared across ⩾3 countries concurrently.
- Cross-border (not just internal) climate migration at scale.
- A maritime/cable security incident coinciding with the drought peak.

Strategic opportunities

- **United States:** lead HADR and infrastructure-resilience to convert relief into durable alignment; friend-shore minerals and semiconductors; champion open food trade and Mekong transparency.
- **China:** position as crisis financier/relief provider and water-cooperation partner (double-edged — risks backlash).
- **Regional governments:** use the warning window for anticipatory action; deepen ASEAN food/energy coordination; diversify creditors.
- **Private sector:** adaptation tech (drought-resistant seed, water reuse, grid storage), parametric climate insurance, resilient data-center/semiconductor design, LEO-satellite communications backup.

Forecast 2026–2040

The central expectation (Scenario 2, ~50%) is **chronic, compounding instability** — recurring ENSO-driven crises producing slow deterioration in the most fragile states, steady erosion of fiscal space, persistent distress migration, and gradual Chinese influence accretion checked by ASEAN hedging and allied counter-balancing. Resilience (Scenario 1) and cascading crisis (Scenario 3) sit at roughly ~25% each, with the cascading tail growing more probable toward the late 2030s as climate change intensifies ENSO extremes (IPCC) and as debt and demographic pressures accumulate. The decisive variable is **governance and adaptation investment in the warning windows** — the difference between Scenario 2 and Scenario 3 is largely a function of whether the region uses its forecast lead time.